

1_FinalReport

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Introduction

Data for this analysis were obtained from the Multicenter AIDS Cohort Study (MACS), an ongoing prospective cohort study of the natural and treated histories of HIV-1 infection among homosexual and bisexual men across four major U.S. cities. The analytic dataset includes longitudinal laboratory and quality of life measures collected annually for up to eight years following initiation of highly active antiretroviral therapy (HAART), along with baseline demographic and health information. Year 0 data represent each subject's last untreated visit immediately prior to HAART initiation, with Years 1 through 8 representing on-treatment follow-up visits.

The scientific hypothesis of interest is whether treatment response two years after HAART initiation differs between men who reported hard drug use at baseline and those who did not. This hypothesis is motivated by laboratory and animal evidence suggesting that hard drug use may inhibit immune function and promote HIV replication, though human evidence has been inconclusive. Understanding whether hard drug users have poorer treatment response has direct clinical implications, as it may support more aggressive treatment strategies or targeted referral to drug rehabilitation programs for this population.

This hypothesis was evaluated across four measures of treatment response at Year 2: log10-transformed plasma viral load, CD4+ T cell count, and aggregate physical and mental quality of life scores. Statistically, we tested whether baseline hard drug use was independently associated with each Year 2 outcome after adjusting for potential confounders, using both frequentist linear regression and Bayesian linear regression frameworks. Additionally, we examined whether any observed differences in treatment response between hard drug users and non-users were mediated through differential medication adherence.

Methods

Data Management

Data was obtained from the MACS and included annual visits from Year 0 (last pre-treatment visit) through Year 8 for HIV-positive men who initiated HAART. The analytic dataset was constructed by linking each subject's Year 0 baseline visit to their Year 2 follow-up visit, retaining baseline covariate values and Year 2 outcome measures for a single cross-sectional analysis of treatment response. Subjects who did not have both a Year 0 and Year 2 visit were excluded. Additionally, subjects missing any outcome or covariate value used in the analysis were excluded via complete case analysis, resulting in a final analytic sample of 463 subjects.

BMI values recorded as negative or exceeding 250, were identified as biologically implausible and set to missing prior to analysis. Hard drug use was defined at baseline (Year 0) only, consistent with the scientific question of interest, and was treated as a time-fixed binary exposure. Several covariates were collapsed into binary categories to ensure adequate cell counts and improve interpretability. Smoking status was collapsed into current smoker versus not current smoker, combining never and former smokers into a single reference

category. Education was collapsed as having obtained at least a four-year college degree versus less than a four-year degree. Race and ethnicity was collapsed into Non-Hispanic White versus all other racial and ethnic groups. Medication adherence was categorized as greater than 95% adherent, combining the 100% and 95–99% adherence categories, versus 95% or less adherent, combining the 75–94% and less than 75% categories. This 95% threshold is a standard clinical cutoff for HAART adherence.

Preliminary Data Analysis

Distributions of all four outcome variables were examined at Year 2 using histograms, density plots, and quantile-quantile plots to assess normality and inform transformation decisions. Plasma viral load was log10-transformed due to strong right skew, consistent with standard practice in HIV research. Physical and mental quality of life scores were reflected and log-transformed using the transformation $\log(101 - \text{score})$ to address left skew and bounded distributions, where higher transformed values indicate worse quality of life. CD4+ T cell count was retained on its original scale after inspection suggested adequate distributional properties for linear regression. Baseline values of each outcome were similarly transformed to serve as covariates in the regression models. Descriptive statistics were computed for all analytic variables overall and stratified by baseline hard drug use status.

Analysis

The association between baseline hard drug use and each Year 2 outcome was estimated using multiple linear regression. Four separate models were fit, one for each outcome: log10 viral load, CD4+ T cell count, reflected log physical quality of life, and reflected log mental quality of life. In each model, the primary explanatory variable was baseline hard drug use (hard drug user versus no hard drug use, the reference group). Each model was adjusted for the following covariates selected a priori based on clinical and epidemiological knowledge: baseline value of the outcome (to account for pre-treatment differences), age, BMI, smoking status, education, and race/ethnicity. The coefficient on hard drug use in each model represented the adjusted difference in the Year 2 outcome between hard drug users and non-users, holding all other covariates constant. Statistical significance was assessed at the $\alpha = 0.05$ level using two-sided tests. No adjustment for multiple comparisons was made across the four outcomes, as each outcome represents a distinct dimension of treatment response and the analyses were considered exploratory in nature.

To evaluate whether medication adherence might explain any observed differences in treatment response between hard drug users and non-users, each model was fit twice: once without adherence as a covariate and once with adherence included. Adherence was considered a potential mediator of the relationship between hard drug use and treatment outcomes, as hard drug use may reduce adherence, which in turn could worsen virologic and immunologic response. A formal mediation analysis was subsequently conducted using the potential outcomes framework to decompose the total effect of hard drug use on each outcome into an average direct effect and an average causal mediation effect operating through adherence. The mediator model was a logistic regression predicting poor adherence ($\geq 95\%$) from baseline hard drug use and all covariates. A quasi-Bayesian approximation with 1,000 simulations was used to obtain confidence intervals for mediation quantities.

All analyses were conducted in parallel using both frequentist and Bayesian linear regression frameworks for comparison. Bayesian models used the same outcome, predictor, and covariate structure as the frequentist models. Weakly informative priors were specified for all regression coefficients: a normal prior with mean zero and standard deviation of 2 on the log scale for viral load and quality of life outcomes, and standard deviation of 100 for CD4 count, reflecting the larger scale of that outcome. A half-normal prior with standard deviation of 5 was placed on the residual standard deviation. Each Bayesian model was fit using Hamiltonian Monte Carlo sampling via Stan, with 4 chains of 1,000 warmup and 1,000 sampling iterations each, yielding 4,000 post-warmup draws. Convergence was assessed using the Gelman-Rubin diagnostic ($\hat{R}$), with values below 1.01 indicating adequate convergence, and effective sample size, with values above 1,000 considered sufficient. Posterior summaries are reported as posterior means with 95% highest posterior

density intervals. Bayesian mediation was conducted by combining posterior draws from the separately fit logistic mediator and linear outcome models to obtain full posterior distributions for the average causal mediation effect, average direct effect, total effect, and proportion mediated. Model fit was assessed using the widely applicable information criterion and leave-one-out cross-validation.

Results

Study Sample

Of the 463 subjects included in the analytic sample, 427 (92.2%) reported no hard drug use at baseline and 36 (7.8%) reported hard drug use at baseline. Baseline characteristics are presented in Table 1. The two groups were similar in age (mean 43.1 years in non-users versus 43.9 years in hard drug users) and BMI (25.3 versus 23.6). Hard drug users were more likely to be current smokers (75.0% versus 34.9%) and less likely to have obtained a four-year college degree (27.8% versus 44.7%) compared to non-users. Hard drug users were also less likely to identify as Non-Hispanic White (52.8% versus 64.9%).

Table 1: Baseline Characteristics of Analytic Sample by Hard Drug Use at Baseline. Mean (SD) for continuous variables; n (%) for categorical variables.

Characteristic	Overall (N=463)	No Hard Drugs (N=427)	Hard Drug User (N=36)
Age (years)	43.2 (8.7)	43.1 (8.7)	43.9 (9.5)
BMI (kg/m ²)	25.2 (4.3)	25.3 (4.4)	23.6 (3.4)
Race/Ethnicity			
Non-Hispanic White	296 (63.9%)	277 (64.9%)	19 (52.8%)
Other	167 (36.1%)	150 (35.1%)	17 (47.2%)
Education			
≥4 Years College	201 (43.4%)	191 (44.7%)	10 (27.8%)
<4 Years College	262 (56.6%)	236 (55.3%)	26 (72.2%)
Smoking Status			
Current Smoker	176 (38%)	149 (34.9%)	27 (75%)
Not Current Smoker	287 (62%)	278 (65.1%)	9 (25%)

Frequentist Results

Results from the frequentist linear regression models are presented in Table 2. Baseline hard drug use was not significantly associated with Year 2 log₁₀ viral load after adjusting for baseline viral load and covariates ($\beta = -0.059$, 95% CI: -0.457 to 0.339 , $p = 0.770$). This difference does not meet the pre-specified clinically meaningful threshold of a 0.5 log₁₀ change in viral load.

Baseline hard drug use was significantly associated with lower Year 2 CD4⁺ T cell count ($\beta = -163.880$, 95% CI: -226.345 to -101.415 , $p < 0.001$), indicating that hard drug users had approximately 164 fewer CD4⁺ T cells at Year 2 compared to non-users after adjustment. This difference exceeds the pre-specified clinically meaningful threshold of 50 cells.

Table 2: Effect of Baseline Hard Drug Use on Year 2 Outcomes:
With and Without Adherence Adjustment

Outcome	Estimate	SE	p	95% CI
Viral Load (log10)				
Viral Load (log10) - No ADH	-0.059	0.202	0.7698	(-0.457, 0.339)
Viral Load (log10) - With ADH	-0.013	0.202	0.95	(-0.41, 0.385)
CD4 Count				
CD4 Count - No ADH	-163.880	31.786	<0.001	(-226.345, -101.415)
CD4 Count - With ADH	-169.683	31.819	<0.001	(-232.215, -107.151)
Physical Quality of Life (reflected log)				
Physical QoL (refl log) - No ADH	0.054	0.025	0.0315	(0.005, 0.104)
Physical QoL (refl log) - With ADH	0.059	0.025	0.0203	(0.009, 0.108)
Mental Quality of Life (reflected log)				
Mental QoL (refl log) - No ADH	-0.008	0.030	0.7838	(-0.067, 0.051)
Mental QoL (refl log) - With ADH	-0.004	0.030	0.8837	(-0.064, 0.055)

Hard drug use at baseline was significantly associated with worse physical quality of life at Year 2 on the reflected log scale ($\beta = 0.054$, 95% CI: 0.005 to 0.104, $p = 0.032$). Because the outcome is reflected, a positive coefficient indicates worse physical quality of life in hard drug users relative to non-users. Hard drug use was not significantly associated with mental quality of life ($\beta = -0.008$, 95% CI: -0.067 to 0.051 , $p = 0.784$).

Adherence Adjustment

When medication adherence was added to each model, the coefficients for hard drug use changed minimally across all four outcomes (Table 2), suggesting that adherence did not substantially confound or mediate the relationship between hard drug use and treatment response in the unadjusted models. The adherence-adjusted estimates for CD4 count ($\beta = -169.683$, 95% CI: -232.215 to -107.151 , $p < 0.001$) and physical quality of life ($\beta = 0.059$, 95% CI: 0.009 to 0.108 , $p = 0.020$) remained statistically significant and were of similar magnitude to the unadjusted estimates.

Bayesian Results

Bayesian linear regression results are presented in Table 3. All models demonstrated adequate convergence, with R-hat values at or below 1.001 and effective sample sizes exceeding 3,000 for all parameters. Results were highly consistent with the frequentist analyses. The posterior mean for the hard drug use coefficient for CD4 count was -164.842 (95% HPDI: -208.090 to -123.422), with a posterior probability of 1.000 that the effect was negative, providing strong evidence of a harmful effect of hard drug use on CD4 recovery. For physical quality of life, the posterior mean was 0.054 (95% HPDI: 0.007 to 0.104) with a posterior directional probability of 0.986. Viral load (posterior mean: -0.059 , 95% HPDI: -0.465 to 0.312 , $P(\text{direction}) = 0.613$) and mental quality of life (posterior mean: -0.008 , 95% HPDI: -0.068 to 0.048 , $P(\text{direction}) = 0.605$) showed no meaningful evidence of an association with baseline hard drug use. Bayesian mediation results (Table 5) were consistent with the frequentist mediation findings, with the indirect effect through adherence representing a small and uncertain proportion of the total effect across all outcomes.

Table 3: Bayesian Results: Effect of Baseline Hard Drug Use on Year 2 Outcomes, With and Without Adherence Adjustment

Outcome	Estimate	Posterior SD	95% HPDI	P(Direction)	ESS	Rhat
Viral Load (log10)						
Viral Load (log10) - No ADH	-0.059	0.199	(-0.465, 0.312)	0.613	4067	1.000
Viral Load (log10) - With ADH	-0.015	0.199	(-0.39, 0.39)	0.532	4438	1.001
CD4 Count						
CD4 Count - No ADH	-164.842	21.648	(-208.09, -123.422)	1.000	5143	1.000
CD4 Count - With ADH	-171.682	21.047	(-211.394, -128.849)	1.000	5571	1.000
Physical QoL (reflected log)						
Physical QoL (refl log) - No ADH	0.054	0.025	(0.007, 0.104)	0.986	3055	1.000
Physical QoL (refl log) - With ADH	0.058	0.025	(0.009, 0.105)	0.990	3226	1.000
Mental QoL (reflected log)						
Mental QoL (refl log) - No ADH	-0.008	0.030	(-0.068, 0.048)	0.605	3335	1.005
Mental QoL (refl log) - With ADH	-0.004	0.031	(-0.064, 0.056)	0.556	4410	1.000

Note:

Estimate = posterior mean. Posterior SD = posterior standard deviation.

HPDI = 95% Highest Posterior Density Interval.

P(Direction) = posterior probability that the effect is in the direction of the estimate.

ESS = Effective Sample Size; values > 1000 indicate adequate sampling.

Rhat < 1.01 indicates chain convergence.

For reflected log QoL outcomes, positive estimates indicate worse quality of life.

Hard drug users vs. non-users at baseline.

Mediation Analysis

Formal mediation analysis confirmed that adherence did not meaningfully mediate the effect of hard drug use on any outcome (Table 4). For CD4 count, the average causal mediation effect through adherence was 4.194 cells/mm³ (95% CI: -2.913 to 11.652, $p = 0.170$), representing only -2.5% of the total effect, with the large majority of the effect operating through direct pathways. For physical quality of life, the indirect effect through adherence was -0.003 (95% CI: -0.009 to 0.002, $p = 0.184$), representing -5.6% of the total effect. Neither the viral load nor mental quality of life mediation effects were statistically significant.

Table 4: Mediation Analysis: Effect of Baseline Hard Drug Use on Year 2 Outcomes Mediated Through Medication Adherence

Outcome	Indirect (ACME)			Direct (ADE)			Total	
	Estimate	95% CI	p	Estimate	95% CI	p	Estimate	95% CI
Viral Load (log10)	-0.033	(-0.083, 0.022)	0.142	-0.009	(-0.393, 0.399)	0.942	-0.043	(-0.436, 0.350)
CD4 Count	4.194	(-2.913, 11.652)	0.17	-169.149	(-229.428, -104.81)	<0.001	-164.955	(-225.141, -104.769)
Physical QoL (refl log)	-0.003	(-0.009, 0.002)	0.184	0.059	(0.011, 0.109)	0.014	0.056	(0.000, 0.112)
Mental QoL (refl log)	-0.003	(-0.009, 0.003)	0.292	-0.004	(-0.062, 0.057)	0.854	-0.007	(-0.065, 0.051)

Note:

ACME = Average Causal Mediation Effect (indirect path through adherence).

ADE = Average Direct Effect (path not through adherence).

Prop. Mediated = ACME / Total Effect.

CI's based on quasi-Bayesian approximation with 1000 simulations.

Quasi-Bayesian method used due to sparse cell counts in bootstrapped resamples.
For reflected log QoL outcomes, positive estimates indicate worse quality of life.
Hard drug users vs. non-users at baseline.

Table 5: Bayesian Mediation Analysis: Effect of Baseline Hard Drug Use on Year 2 Outcomes Mediated Through Medication Adherence

Outcome	Indirect (ACME)			Direct (ADE)			Total (TE)	
	Estimate	95% HPDI	P(>0)	Estimate	95% HPDI	P(>0)	Estimate	95% HPDI
Viral Load (log10)	-0.030	(-0.063, 0)	0.020	-0.015	(-0.39, 0.39)	0.468	-0.045	(-0.4, 0.3)
CD4 Count	4.658	(0.748, 9.153)	0.986	-171.682	(-211.394, -128.849)	0.000	-167.025	(-207.5, -126.5)
Physical QoL (refl log)	-0.003	(-0.007, 0)	0.044	0.058	(0.009, 0.105)	0.990	0.055	(0.006, 0.104)
Mental QoL (refl log)	-0.002	(-0.007, 0.001)	0.092	-0.004	(-0.064, 0.056)	0.444	-0.007	(-0.067, 0.053)

Note:

ACME = Average Causal Mediation Effect (indirect path through adherence).

ADE = Average Direct Effect (path not through adherence).

P(>0) = posterior probability that the effect is positive.

Prop. Mediated = posterior mean of ACME / Total Effect.

Mediation quantities computed by combining posterior draws from separate logistic (mediator) and linear (outcome) Stan models.

For reflected log QoL outcomes, positive estimates indicate worse quality of life.

Hard drug users vs. non-users at baseline.

Discussion/Conclusion

This analysis examined whether treatment response two years after HAART initiation differed between HIV-positive men who reported hard drug use at baseline and those who did not, across four outcomes: viral load, CD4+ T cell count, physical quality of life, and mental quality of life. After adjustment for baseline outcome values, age, BMI, smoking status, education, and race/ethnicity, hard drug users demonstrated significantly lower CD4+ T cell counts and worse physical quality of life at Year 2 compared to non-users. These differences were clinically meaningful, with hard drug users averaging approximately 164 fewer CD4+ T cells, well exceeding the 50-cell clinical threshold, and demonstrating worse physical quality of life on the reflected log scale. No significant differences were observed for viral load or mental quality of life. Results were highly consistent between the frequentist and Bayesian frameworks, lending additional confidence to these findings.

The observation that hard drug use was associated with impaired CD4 recovery but not with viral load suppression is clinically interesting. It suggests that while hard drug users may achieve comparable virologic suppression under HAART, their immune reconstitution remains compromised, consistent with laboratory evidence that hard drugs may directly inhibit immune function independent of viral replication. The significant association with physical but not mental quality of life may reflect the direct physiological consequences of hard drug use on physical health, while mental health trajectories may be driven by other factors not captured in this analysis.

Contrary to our hypothesis, medication adherence did not appear to mediate the relationship between hard drug use and treatment outcomes. Both frequentist and Bayesian mediation analyses found that the indirect effect through adherence was small and not statistically significant for any outcome, with the vast majority of the total effect operating through direct pathways. This suggests that the poorer immunologic and physical outcomes observed in hard drug users are not simply a consequence of missed doses, and that interventions

focused solely on improving adherence may be insufficient to close the gap in treatment response between hard drug users and non-users.

Several limitations of this analysis warrant consideration. First, the hard drug user group was small, comprising only 36 of 463 subjects (7.8%), which substantially limits statistical power to detect moderate associations and increases the width of confidence intervals, particularly for viral load and quality of life outcomes. The investigator should interpret non-significant findings with caution, as they may reflect insufficient power rather than a true absence of effect. Second, hard drug use was defined only at baseline, and subjects may have changed their drug use behavior over the two-year follow-up period. Misclassification of exposure over time could bias estimates toward the null. Third, the analysis relied on complete case analysis, excluding subjects with missing data on any outcome or covariate, which could introduce bias if missingness was related to drug use status or health trajectory. Fourth, the MACS cohort consists exclusively of homosexual and bisexual men, limiting generalizability to other populations affected by HIV, including women and heterosexual men. Finally, while we adjusted for several important confounders, residual confounding from unmeasured variables such as other substance use, mental health diagnoses, and social support cannot be ruled out.

These findings suggest that baseline hard drug use is associated with meaningful deficits in immunologic recovery and physical quality of life two years after HAART initiation, independent of medication adherence. Clinicians treating HIV-positive patients who use hard drugs may need to consider more aggressive immunologic monitoring and targeted interventions beyond adherence support to improve treatment outcomes in this population.

Reproducible Research

GitHub Link: <https://github.com/asteinwand/BIOS6624.git>

File Path to data: `hivdat <- read.csv("C:/Users/stein/OneDrive/Documents/School/2026 Spring/Advanced Methods/BIOS6624/Project 1/Data Raw/hiv_6624_final.csv")`